

Detection Engineering for Data Loss Prevention: Designing Splunk-Based Alerts and Response Plans for Data Exfiltration Incidents

Pedram Shahbazi, Robert (Ash) Drake

Project (CS-6399-D10)
Dr. Erdoğan Doğdu, Dr. Roya Choupani

Progress Report # 5

**Master of Computer Science
Department of Computer Science
Angelo State University
Spring 2026**

Executive Summary

This progress report documents the completion of all project phases for the Detection Engineering for Data Loss Prevention project. Over the course of fourteen weeks, a structured Splunk-based detection-engineering framework was designed, implemented, tested, and evaluated for four distinct data-exfiltration vectors: high-volume outbound transfers, unsanctioned cloud storage access, DNS-based exfiltration, and email-based exfiltration.

All functional and non-functional requirements defined in the original project proposal have been met. Four production-ready SPL detection rules, three supporting lookup tables, one reusable SPL macro, and four incident response playbooks have been finalized and committed to version control. Comprehensive testing demonstrated 100% recall across all simulated exfiltration scenarios and 94% overall precision, with the single false positive fully understood and mitigated through an operational exception mechanism.

The project deliverable package — comprising all SPL artefacts, lookup tables, playbooks, the synthetic test dataset, evaluation results — has been assembled and will be submitted. This report summarizes the work completed in Phase 6 (documentation and finalization), presents the complete system architecture and evaluation results, and discusses lessons learned, limitations, and directions for future work.

Problem Statement

Data exfiltration incidents, whether driven by external attackers or malicious insiders, represent a critical threat to the confidentiality and integrity of organizational information assets. As enterprises increasingly adopt cloud services, remote work models, and heterogeneous infrastructures, data can leave corporate boundaries through a wide variety of channels, including web uploads, cloud storage platforms, email, collaboration tools, and covert network protocols. Traditional Data Loss Prevention (DLP) solutions typically focus on content inspection and policy enforcement within a single channel, which often leads to fragmented visibility and an overwhelming number of isolated alerts.

Security Information and Event Management (SIEM) platforms such as Splunk are designed to aggregate logs and events from multiple security and IT systems, offering a centralized view of activity across the environment. However, many organizations underutilize this aggregated data for DLP-specific detection engineering and lack well-defined response processes tailored to data exfiltration scenarios. The core problem addressed by this project is the absence of a structured, Splunk-based framework that uses aggregated telemetry to detect diverse data-exfiltration behaviors and couples these detections with clear, actionable incident response plans.

This project designed, implemented, and evaluated a detection-engineering solution for DLP scenarios operating on aggregated data within Splunk. The solution specified detection use cases for multiple exfiltration vectors, defined the required data and system architecture, implemented correlation and alerting logic, and documented

response playbooks guiding SOC analysts from alert generation through to containment and recovery.

Related Work

Literature References

Prior research on data exfiltration has explored both signature-based and behavior-based detection approaches. Network-centric studies analyze flow statistics, traffic volumes, and protocol-specific features to identify anomalous outbound behavior indicative of data theft. Host-centric and application-centric works focus on access patterns, file operations, and process behaviors. More recent research has applied statistical models and machine learning to detect deviations from established baselines, particularly for low-and-slow exfiltration attempts that evade simple volume thresholds.

In the domain of security operations, literature on SIEM and SOC efficiency emphasizes the importance of correlation across heterogeneous data sources and the codification of response procedures into structured playbooks. These works highlight that effective detection is not only a matter of signal quality but also of how well alerts integrate into the operational workflows of analysts, including triage, enrichment, and remediation steps.

Other Products and Projects

Commercial DLP solutions and Cloud Access Security Brokers (CASBs) provide policy-based capabilities for monitoring data movement across endpoints, email, web gateways, and cloud services. SIEM and XDR platforms offer predefined rules for suspicious outbound traffic, abnormal cloud activity, and DNS tunneling, but require customization for specific environments. This project addresses the gap by engineering detections directly within Splunk using a structured, use-case-driven methodology and coupling each detection with a concrete response playbook.

Critical Comparison with the Proposed Design

The project distinguishes itself by treating DLP for data exfiltration as an end-to-end detection-engineering problem within Splunk, spanning multiple exfiltration channels through aggregated analytics, and tightly coupling each detection rule with an operational incident response playbook. This approach produces artefacts directly consumable by SOC teams, rather than generic guidance requiring significant adaptation.

Requirements — Final Status

Functional Requirements

Requirement	Description	Status
FR1 – Data Ingestion	Ingest network/proxy, DNS, email audit, and DLP logs into Splunk with field normalization.	Complete
FR2 – Detection Use Cases	Define and implement at least four exfiltration detection use cases.	Complete – 4 of 4 implemented
FR3 – Detection Logic	SPL searches aggregating per-entity metrics and generating contextual alerts.	Complete – all rules finalized
FR4 – Response Playbooks	Four-step playbooks for each use case covering validation, enrichment, containment, and escalation.	Complete – all 4 playbooks finalized
FR5 – Evaluation	Controlled tests using synthetic data; false-positive assessment; structured results report.	Complete – full test plan executed

Non-Functional Requirements

Requirement	Target	Achieved
Performance	Queries complete within acceptable operational timeframes.	Yes – max 14.1 s for 24-hr DNS window
Scalability	Efficient aggregations viable as data volumes grow.	Yes – parameterized macros and summary indexes recommended
Reliability	Consistent detection with documented error-handling.	Yes – all rules tested; missing-field handling documented
Explainability	Transparent logic, thresholds, and limitations documented.	Yes – all rules include inline rationale and tuning notes
Maintainability	Modular, version-controlled rules and playbooks.	Yes – Git repository with all artefacts
Compliance & Privacy	Anonymized or synthetic data used throughout.	Yes – no real production data used
Ingestion Volume	Under 500 MB/day (license compliance).	Yes – synthetic dataset within limit
Uptime / Delay	100% uptime, <5 min ingestion delay.	Yes – lab environment met target

Final System Architecture

The completed system is organized into four layers, each implemented and validated over the course of the project.

Layer 1 – Data Source Layer

The data source layer comprises the log-producing components whose events are ingested into Splunk. For this project, all data originated from the synthetic dataset; in a production deployment these would be replaced by live feeds from the corresponding infrastructure.

Source Type	Index	Key Fields	Use Cases Served
Network/Proxy Logs	dlp_test	src_ip, dest_ip, domain, bytes, user	UC1, UC2
DNS Query Logs	dns_logs	src_ip, query_name, query_type, response	UC3
Email Audit Logs	email_logs	user, recipient, subject, action, attachment_size	UC4
DLP Policy Logs	dlp_policy	user, host, policy_name, action, severity	Enrichment (all)

Layer 2 – Ingestion and Normalization Layer

Raw logs are ingested into dedicated Splunk indexes and assigned source types that drive field extraction. A common set of normalized fields — timestamp, user, host, src_ip, dest_ip, domain, bytes, and action — enables cross-source correlation. Lookup tables provide entity enrichment: user-to-department mapping, asset classification, and domain sanctioning status.

Layer 3 – Analytics and Detection Layer

The analytics layer implements five SPL artefacts (four detection rules and one reusable macro) that transform raw events into candidate exfiltration indicators. Each rule aggregates data per entity over a configurable time window, computes one or more discriminating features, and filters results against defined thresholds. The DNS rule additionally calls the calc_entropy macro to compute Shannon entropy of subdomain strings.

Layer 4 – Alerting and Response Layer

When a detection rule fires, it generates a Splunk alert or notable event enriched with the involved entities, the supporting evidence, and a severity classification. Each alert entry point maps to one of the four documented incident response playbooks, which

guide the SOC analyst through triage, enrichment, containment, and escalation. In the current implementation, playbook execution is semi-manual; the playbook structure is designed to support future automation via a SOAR platform.

Complete Detection Framework

Summary of All Detection Use Cases

Use Case	Vector	Primary Signal	Threshold	SPL Artefact
UC1	High-Volume Outbound	Total bytes per src_ip per domain	> 500 MB / window	uc1_high_volume.spl
UC2	Unsanctioned Domain	Domain match in unsanctioned_domains.csv	Any match + volume	uc2_unsanctioned.spl
UC3	DNS Exfiltration	Query count + subdomain length + entropy	Count>300, len>40, entropy>3.5	uc3_dns_exfil.spl
UC4	Email Exfiltration	External sends / attachment volume	>50 sends OR >100 MB attachments	uc4_email_exfil.spl

UC1 – High-Volume Outbound Transfer (Final SPL)

```
index=dlp_test
| stats sum(bytes) as total_bytes by src_ip domain
| where total_bytes > `outbound_bytes_threshold`
| lookup asset_inventory src_ip OUTPUT hostname department
| lookup domain_classification domain OUTPUT category sanctioned
| sort - total_bytes
```

UC2 – Unsanctioned Cloud Storage Access (Final SPL)

```
index=dlp_test
| lookup unsanctioned_domains domain OUTPUT domain as flagged_domain
| where isnotnull(flagged_domain)
| stats sum(bytes) as total_bytes count as events by src_ip domain
| lookup asset_inventory src_ip OUTPUT hostname department
| sort - total_bytes
```

UC3 – DNS-Based Exfiltration (Final SPL with Entropy)

```
index=dns_logs
| eval subdomain=mvindex(split(query_name,"."),0)
| eval sub_len=len(subdomain)
| eval entropy=`calc_entropy(subdomain)`
```

```
| stats count as query_count avg(sub_len) as avg_len avg(entropy) as
avg_entropy
  by src_ip
| where query_count > `dns_query_count_threshold`
  AND avg_len > `dns_subdomain_len_threshold`
  AND avg_entropy > `dns_entropy_threshold`
| lookup asset_inventory src_ip OUTPUT hostname department
| sort - avg_entropy
```

UC4 – Email-Based Exfiltration (Final SPL)

```
index=email_logs (action=send OR action=forward)
| where NOT match(recipient,"@company\.com")
| lookup authorized_senders user OUTPUT user as exempt_user
| where isnull(exempt_user)
| stats count as external_sends dc(recipient) as unique_recipients
  sum(attachment_size) as total_attach_bytes by user
| where external_sends > `email_send_threshold`
  OR total_attach_bytes > `email_attach_bytes_threshold`
| eval total_attach_mb=round(total_attach_bytes/1048576,1)
| sort - external_sends
```

Complete Evaluation Results

Detection Performance — Full Test Suite

Use Case	True Positives	False Positives	False Negatives	Precision	Recall	F1
UC1 – High-Volume Outbound	4	0	0	100%	100%	1.00
UC2 – Unsanctioned Domain	6	0	0	100%	100%	1.00
UC3 – DNS Exfiltration	2	0	0	100%	100%	1.00
UC4 – Email Exfiltration	3	1	0	75%	100%	0.86
Overall	15	1	0	94%	100%	0.97

Scenario-Based Test Results

Scenario	Vector(s)	Rules Triggered	Playbook(s) Invoked	Result
S1: Insider uploads archive to Dropbox	UC1 + UC2	Both	PB1, PB2	Pass
S2: Employee bulk-forwards emails pre-resignation	UC4	UC4	PB4	Pass
S3: Compromised host runs DNS tunnel client	UC3	UC3	PB3	Pass
S4: Staged sync to WeTransfer over 3 hours	UC1 + UC2	Both	PB1, PB2	Pass
S5: Mass attachment send to external contacts	UC4	UC4	PB4	Pass

Performance Benchmarks

Detection Rule	1-Hour Window	24-Hour Window	Meets SLA?	Optimization Note
UC1 – High-Volume Outbound	1.4 s	8.2 s	Yes	None required
UC2 – Unsanctioned Domain	1.1 s	6.7 s	Yes	None required
UC3 – DNS Exfiltration	2.3 s	14.1 s	Yes	Pre-compute entropy at index time for production
UC4 – Email Exfiltration	1.6 s	9.4 s	Yes	None required

Qualitative Operational Assessment

A structured analyst walkthrough was conducted for each of the four playbooks using the alert outputs generated during scenario-based testing. The following observations were recorded:

- Alert fidelity: Each alert contained sufficient context — user identity, source host, destination, channel, and supporting event count — to allow an analyst to form an initial hypothesis without issuing additional Splunk queries.

- **Playbook completeness:** All four playbooks were found to cover the expected triage and containment actions for their respective scenarios. No gaps in decision logic were identified during the walkthrough.
- **Threshold documentation:** All thresholds are expressed as named macros with accompanying documentation files explaining their derivation, the behavioral baseline they reference, and the conditions under which recalibration is recommended.
- **Lookup table maintainability:** The three CSV lookup tables (unsanctioned_domains, authorized_senders, sanctioned_services) will be stored in the Git repository.
- **SPL modularity:** The calc_entropy macro and threshold macros allow tuning and extension of detection logic without modifying core rule files, reducing the risk of introducing syntax errors during routine updates.

Incident Response Playbooks — Final Summary

All four incident response playbooks have been finalized. The table below provides a condensed reference view of each playbook's key actions. Full playbook documentation, including decision trees and escalation matrices, is included in the project deliverable package.

Playbook	Use Case	Key Containment Action	Escalation Trigger
PB1	High-Volume Outbound	Block destination at web proxy/firewall; optional host isolation	Confirmed sensitive data type (PII, IP, financial)
PB2	Unsanctioned Domain	Block domain via proxy policy; initiate user interview	Repeated violations or high data volume
PB3	DNS Exfiltration	Block domain at DNS resolver; isolate host; capture forensic image	Any confirmed DNS tunnel – immediate escalation
PB4	Email Exfiltration	Disable forwarding rule; legal hold on mailbox; block external email	HR coordination required for insider-threat cases

Finalized Material Inventory

All project materials have been completed, reviewed, version-controlled and will be assembled into the deliverable package. The following table provides the complete inventory.

Artefact	Type	Description	Status
uc1_high_volume.spl	SPL Rule	Parameterized detection for outbound transfer volume	Finalized
uc2_unsanctioned.spl	SPL Rule	Lookup-based detection for unsanctioned domain access	Finalized
uc3_dns_exfil.spl	SPL Rule	Composite DNS tunneling detection with entropy	Finalized
uc4_email_exfil.spl	SPL Rule	Dual-rule email exfiltration detection	Finalized
calc_entropy.spl	SPL Macro	Reusable Shannon entropy computation macro	Finalized
thresholds.conf	Config	Named macro definitions for all tunable thresholds	Finalized
unsanctioned_domains.csv	Lookup Table	List of flagged cloud storage and file-sharing domains	Finalized
authorized_senders.csv	Lookup Table	Approved bulk-email senders exempt from UC4	Finalized
sanctioned_services.csv	Lookup Table	Approved SaaS services for UC2 exception handling	Finalized
asset_inventory.csv	Lookup Table	Host-to-user-to-department mapping	Finalized
playbook_pb1.md	Playbook	4-step playbook for high-volume outbound alerts	Finalized
playbook_pb2.md	Playbook	4-step playbook for unsanctioned domain alerts	Finalized
playbook_pb3.md	Playbook	4-step playbook for DNS exfiltration alerts	Finalized
playbook_pb4.md	Playbook	4-step playbook for email exfiltration alerts	Finalized
synthetic_dataset/	Test Data	25,000-event multi-source synthetic log dataset	Finalized
benign_dataset/	Test Data	8,000-event benign-only validation dataset	Finalized

Artefact	Type	Description	Status
test_catalog.xlsx	Test Artefact	18 test cases with inputs, expected, and actual results	Finalized
tuning_log.md	Test Artefact	Chronological log of all threshold adjustments and impact	Finalized
eval_results.md	Evaluation	Full quantitative and qualitative evaluation results	Finalized

Phase 6 — Documentation and Finalization

Phase 6 represents the final project phase, covering Weeks 12–14. The work completed in this phase is described below.

Final Report Preparation

The final report was drafted iteratively across the reporting period. Sections were reviewed against the original project proposal and requirements to confirm all commitments had been addressed. The report structure follows the project's logical progression: problem statement and related work, requirements, system design, implementation, evaluation, and conclusion.

- Architecture section: The four-layer system architecture was documented with descriptions of each component, its role, and its relationship to adjacent layers.
- Implementation section: All four detection use cases were documented with final SPL code, threshold rationale, and annotated result tables from testing.
- Evaluation section: Quantitative metrics (precision, recall, F1) and qualitative observations from the analyst walkthrough were compiled and presented.
- Future work section: Directions for extending the framework were identified and documented, covering production deployment, SOAR integration, and machine-learning enhancements.

Risk Plan — Final Resolution

Risk	Original Mitigation	Final Outcome
R1 – Limited Realistic Data	Use synthetic datasets; document limitations.	Managed. A 25,000-event synthetic dataset provided sufficient coverage. Limitations are documented in the evaluation section. Future work identifies use of public capture datasets as an extension.
R2 – High False-Positive Rate	Baselining, per-entity thresholds, iterative tuning.	Resolved. 94% overall precision achieved. The single remaining FP is fully characterized and mitigated via the <code>authorized_senders</code> lookup.
R3 – Time Constraints	Prioritize core four use cases; treat extensions as future work.	Resolved. All four use cases completed on schedule. Additional ideas (ML scoring, SOAR automation) documented as future work.
R4 – Tooling Limitations	Conceptual designs where full integration not possible.	Managed. Playbooks remain semi-manual due to absence of SOAR. Structure is SOAR-ready for future automation.
R5 – Detection Complexity	Favor interpretable logic; use macros and parameterization.	Resolved. All rules use macros for complexity abstraction; documentation is complete.

Lessons Learned

Detection Engineering Process

- Iterative tuning is essential. Initial threshold values derived from dataset inspection required adjustment in every use case. Building tuning into the process from the outset — rather than treating it as a post-implementation step — would reduce rework in future projects.
- Multi-signal detections outperform single-signal rules. The DNS detection rule demonstrated that combining query volume, subdomain length, and entropy produces substantially better precision than any individual signal alone. This principle should guide the design of future detection use cases.
- Parameterization from the start saves time. Encoding thresholds as named macros from the initial implementation made tuning straightforward and prevented the proliferation of hardcoded values across SPL files.

Synthetic Dataset Design

- Realism requires deliberate design. Simply embedding large byte values or high event counts in a dataset does not produce a realistic test environment. Effective synthetic data requires careful modeling of benign behavior distributions so that exfiltration events stand out for the right reasons.
- Benign dataset diversity is as important as malicious coverage. The single false positive in UC4 was only discovered because the benign dataset included a sales team member with a legitimately high email volume. Narrow benign datasets produce overoptimistic precision estimates.

Operational Considerations

- Lookup table maintenance is an ongoing operational commitment. The detection framework's effectiveness depends on lookup tables (unsanctioned domains, authorized senders) being kept current. Organizations adopting this framework should assign ownership of these tables to a named SOC role and establish a review cadence.
- Playbook design benefits from SOC analyst input. The analyst walkthrough conducted in Phase 5 identified several minor clarifications needed in playbook language. In a production environment, involving SOC analysts in playbook design from the outset would improve usability and adoption.
- Version control for SPL is underutilized but highly valuable. Maintaining detection rules and lookup tables in Git provided a clear audit trail of tuning decisions and enabled rollback when threshold changes produced unexpected results. This practice is strongly recommended for production SOC environments.

Limitations

The following limitations apply to the current implementation and should be considered when interpreting results or planning production deployment:

- Synthetic data dependency: All testing was performed on synthetic datasets. While carefully designed, these datasets cannot fully replicate the diversity, noise, and edge cases present in real enterprise environments. Detection performance in production may differ from the results reported here.
- Single-organization scope: The detection thresholds and lookup table contents are calibrated for the synthetic dataset's implied organizational profile. Deploying this framework in a real organization would require recalibration of all thresholds against that organization's baseline traffic patterns.
- Absence of SOAR integration: Playbooks are currently executed semi-manually. Without SOAR automation, response times depend on analyst availability, and steps may be skipped under high alert volume conditions.
- Limited DLP log coverage: DLP policy logs were used primarily for enrichment rather than as primary detection signals. Tighter integration with a commercial DLP solution's alert output could improve detection fidelity, particularly for

content-aware scenarios (e.g., detection of specific sensitive data patterns within transfers).

- No lateral movement correlation: The current framework detects exfiltration-phase activity but does not correlate with earlier attack-lifecycle phases such as credential compromise, privilege escalation, or lateral movement. Integrating these signals could improve early warning capabilities.

Future Work

Several directions for extending and improving the framework are identified below, ordered from highest to lowest implementation priority for a production deployment:

1. SOAR Integration

The most impactful near-term extension is integrating the detection framework with a SOAR platform. Each playbook has been structured with discrete, tool-agnostic action steps that map naturally to SOAR automation blocks. Priority automation candidates include: domain blocking at the proxy/firewall, account suspension notifications to HR systems, and Jira/ServiceNow ticket creation upon alert generation. This would reduce mean time to respond and ensure playbook steps are executed consistently regardless of analyst workload.

2. Baseline Anomaly Detection

The current detection rules use static thresholds calibrated against the synthetic dataset. A production deployment would benefit from dynamic baselines computed using Splunk's machine-learning toolkit (MLTK) or rolling statistical models. Per-entity baselines (e.g., each user's typical daily outbound volume) would allow the detection rules to adapt to seasonal patterns, business cycles, and individual role differences, reducing both false positives and false negatives compared to global static thresholds.

3. Machine Learning Scoring Layer

A risk-scoring layer could be added above the existing rule-based detections, aggregating signals across all four use cases into a composite exfiltration risk score per user or host per day. Users or hosts with scores above a configurable threshold would be surfaced for priority analyst review regardless of which individual rules fired. Candidate algorithms include isolation forest for anomaly scoring and gradient-boosted classifiers trained on labeled exfiltration scenarios.

4. Additional Exfiltration Vectors

The current framework covers four vectors. Additional use cases warranting future implementation include: USB and removable media exfiltration (endpoint DLP log integration), printer-based exfiltration (print audit log analysis), collaboration platform exfiltration (Slack, Teams, SharePoint audit logs), and HTTPS tunneling over non-

standard ports. Each would follow the same detection-engineering methodology applied in this project.

5. Production Data Validation

Testing with real enterprise log data — with appropriate privacy controls and data anonymization — would validate the framework's performance beyond the synthetic dataset and provide an opportunity to recalibrate thresholds against actual organizational baselines. A phased rollout (monitor-only mode before alert mode) is recommended to assess false-positive rates before enabling active alerting.

Implementation Status — Final

Phase	Description	Weeks	Status
Phase 1	Environment Setup and Requirements Realization	1–3	Complete
Phase 2	Data Model and Feature Design	3–4	Complete
Phase 3	Detection Implementation (UC1 & UC2)	5–7	Complete
Phase 4	Detection Implementation (UC3 & UC4) + Playbooks	8–9	Complete
Phase 5	Testing, Tuning, and Evaluation	10–12	Complete
Phase 6	Documentation and Finalization	13–14	Complete

Task Completion Summary

Task	Description	Status
T1	Configure data inputs, indexes, and field extractions	Complete
T2	Define use-case templates and threat models	Complete
T3	Implement feature extraction searches and baselines	Complete
T4	Implement correlation searches for each detection use case	Complete
T5	Draft and refine incident response playbooks	Complete

Task	Description	Status
T6	Develop synthetic data and test scenarios	Complete
T7	Execute tests and collect metrics	Complete
T8	Compile documentation, diagrams, and final report sections	Complete

Project Timeline — Final

Weeks	Milestone	Status
1–2	Requirements finalization and initial Splunk setup	Complete
3	Data ingestion for first core log type; conceptual data model draft	Complete
4–5	Feature extraction searches; UC1 and UC2 detection rules	Complete
6–7	UC3 and UC4 detection rules; initial playbook drafts	Complete
8–9	Playbook finalization; expanded synthetic datasets	Complete
10–11	Full test plan execution; threshold tuning; results documentation	Complete
12–14	Final report; presentation materials; deliverable package assembly	Complete

All project milestones have been completed on schedule. No timeline deviations occurred during the project.

Conclusion

This progress report marks the completion of the Detection Engineering for Data Loss Prevention project. Over fourteen weeks, the project successfully designed, implemented, tested, and evaluated a structured Splunk-based detection-engineering framework addressing four distinct data-exfiltration vectors: high-volume outbound transfers, unsanctioned cloud storage access, DNS-based exfiltration, and email-based exfiltration.

All project requirements — functional and non-functional — have been met. The framework achieved 100% recall across all simulated exfiltration scenarios and 94%

overall precision across more than 33,000 synthetic log events. All four detection rules are production-ready, parameterized, and fully documented. All four incident response playbooks have been finalized and validated through an analyst walkthrough. The complete artefact set has been version-controlled and assembled into the project deliverable package.

The project demonstrates that a rigorous, use-case-driven detection-engineering methodology applied within Splunk can produce high-fidelity, operationally usable DLP detections without reliance on proprietary standalone DLP products. The framework is designed to be maintainable, extensible, and adaptable to real organizational environments. Identified limitations — including synthetic data dependency and the absence of SOAR integration — are fully characterized, and a prioritized roadmap of future enhancements has been documented to guide further development.

This work contributes a practical, end-to-end framework and a complete set of artefacts that SOC teams can directly adopt or adapt to operationalize data-exfiltration detection within a Splunk-based security operations environment.

References

- Splunk Threat Research Team. "Data Exfiltration Detections: Threat Research Release, June 2021." 2021.
- Splunk. "Detecting Data Exfiltration Activities." Splunk Lantern.
- Splunk. "Machine Learning in Security: Detect DNS Data Exfiltration Using Deep Learning." 2022.
- Splunk. "Machine Learning in Splunk Enterprise Security: Unleashing Hidden Threats." 2025.
- SearchInform. "SIEM Workflow Automation: Streamlining Incident Response." 2025.
- Radiant Security. "SOC Playbook Examples for Real-World Cyber Threats." 2026.
- Hamdan, M. "Data Exfiltration Explained: How to Use Splunk and Wireshark to Detect It." 2025.
- Splunk Lantern. "Data Loss Prevention Data." 2025.
- PuppyGraph. "How to Build a SIEM System: Architecture & Tools." 2026.